

# Problem Set 2

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## 1 Measurement

- Quantitative measures used to show insights and policy effects.

## 2 Programming Languages and Visualization

- Python: open source, used for general computing and data science; Matplotlib for visualization.
- R: open source, used for data science; Tidyverse and ggplot2 for visualization.
- Julia: open source, used for general computing and data science, known for its speed; Plots.jl for visualization.
- Tableau: software producing interactive data visualization products.

## 3 Big Data Management Software

- SQL: database language to subset, merge, and perform other common data transformations.
- RDDs: separate large datasets into manageable chunks, designed to withstand disruptions in computing clusters.

## 4 Data Collection

- Web scraping: collecting data directly from the internet via HTML or APIs.
- API: application programming interface to access data.
- Parsing: processing and cleaning collected web data.